ALICE L WALTON SCHOOL OF MEDICINE

Gastrointestinal System: Foundations and Overview

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Adapted from: Gastrointestinal system: Foundations and Framework

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Learning Objectives (3) Versions

After completing this brick, you will be able to:

1. Correlate the organization of the gastrointestinal (GI) tract to its coordinated functions (motility, secretion, digestion, absorption, protection, and elimination).
2. Compare and contrast the regulation of gut function by the autonomic nervous systems (ANS), enteric nervous system (ENS), endocrine hormones, and paracrine factors.
3. Apply knowledge of gastrointestinal physiologic functions to predict how dysfunctions cause common diseases like diarrhea, ulcers, and malabsorption.

CASE CONNECTION

Let's look at 3 different patients: A 28-year-old woman (TL) presents to urgent care with watery diarrhea for 2 days after returning from international travel. A 54-year-old man (RK) with chronic pancreatitis reports greasy, foul-smelling stools. A college student (JM) experiences diarrhea every time she drinks milk. Do all these patients have the same underlying problem? How do normal gut processes prevent these symptoms? Consider your answers as you read, and we'll revisit TL, RK, and JM at the end of the brick.

GO TO CONCLUSION ↓

Overview of the GI Tract Structure and Function (LO1)

The gastrointestinal (GI) tract is best understood not as a collection of separate organs, but as a single, coordinated tube with one overarching mission: move external material through the body, extract useful molecules, keep harmful material out, and maintain fluid and electrolyte balance.

The GI tract is organized as a hollow tube running from mouth to anus, lined throughout by a continuous epithelial surface. This architecture has a profound conceptual implication: the lumen of the GI tract is technically continuous with the outside world. A molecule sitting in the intestinal lumen has not yet entered the body—it only truly crosses into the internal environment at the moment of absorption across the epithelial barrier. This distinction is clinically important because it explains why the gut must perform selective absorption, maintain immune surveillance, and secrete protective substances into what is essentially an external space.

To appreciate how tightly coordinated these functions must be, consider the GI tract through the lens of mass balance: what enters must either be absorbed or excreted. On a typical day, approximately 9 liters of fluid enter the small intestine—roughly 2 liters ingested and 7 liters contributed by GI secretions (saliva, gastric juice, bile, pancreatic fluid, and intestinal secretions). Of this volume, all but approximately 150 mL is reclaimed; the colon delivers the remainder as stool. This means roughly 98% of all fluid that enters the intestinal lumen must be reabsorbed under normal conditions. The same principle governs nutrients: what is ingested minus what is absorbed equals what appears in the stool. This mass balance frame is essential for understanding diarrhea, malabsorption, and steatorrhea—topics we will return to throughout this brick.

Regional Specialization

Each segment of the GI tract is specialized for a distinct role, and understanding this regional map prevents the common mistake of treating the gut as a uniform tube. The table below summarizes the primary function of each major region, with each region shown in Figure 1:

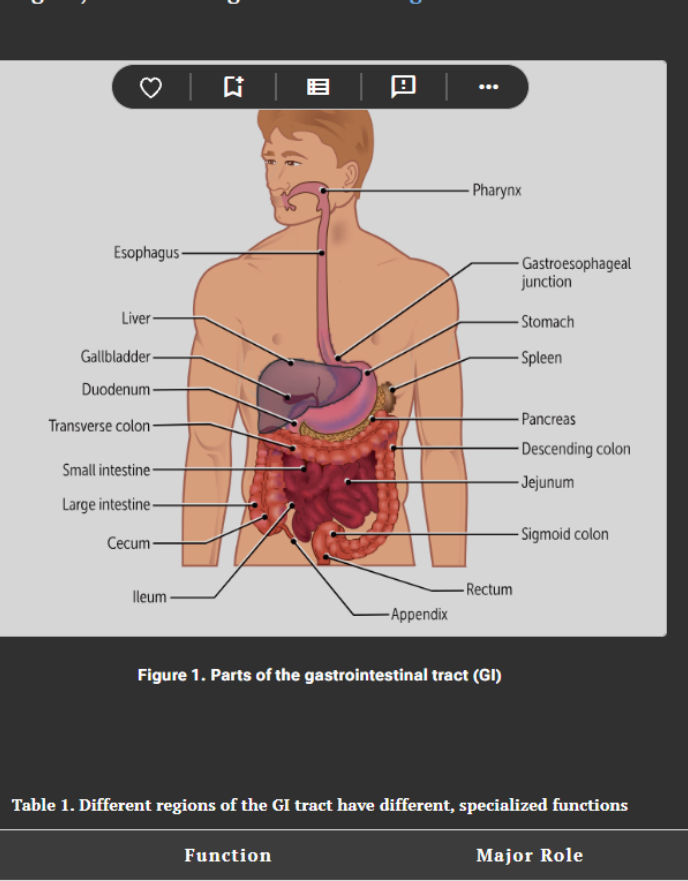

Figure 1. Parts of the gastrointestinal tract (GI)

Table 1. Different regions of the GI tract have different, specialized functions

Region	Function	Major Role
Mouth	Motility, Digestion	Ingest, chew, and initial digestion of carbs and lipids
Esophagus	Motility	Deliver food to stomach
Stomach	Motility, Secretion, Digestion, Protection	Store, acidify, digest protein, regulate emptying. Little absorption occurs.
Duodenum	[illegible]	lize acid; receive bile and eatic enzymes
Jejunum	Absorption, Digestion	Major site of nutrient absorption
Ileum	Absorption	Vitamin B12 and bile salt recovery
Colon	Absorption, Elimination	Water and electrolyte salvage; stool formation
Liver / Gallbladder / Pancreas	Secretion, Digestion, Protection	Provide bile, bicarbonate, and digestive enzymes

Notice that the stomach is not the main site of absorption, a common misconception. The jejunum handles the bulk of nutrient uptake, while the colon's primary job is salvaging water and electrolytes, not absorbing nutrients. The terminal ileum has two highly specific tasks: recovering bile salts for recycling and absorbing vitamin B12. This explains why ileal disease or resection causes predictable deficiencies.

What are the six coordinated functions of the GI tract, and why is the lumen considered outside the body?

Main Functions of the GI System (LO 1)

These six processes are not independent. They are tightly coupled, and disruption of one typically cascades into others. Many GI diseases can be traced back to a failure in one or more of these steps.

Motility and Mixing

Motility refers to the coordinated muscular movements that propel and mix luminal contents. Motility serves two distinct purposes: propulsion of contents away from the mouth and toward the anus, and segmentation (Figure 2).

- Propulsion is accomplished by peristalsis. Peristalsis is the rhythmic contraction of smooth muscle stimulated by enteric nervous signaling and the distension of the gut as the chyme bolus moves down towards the anus.
- Segmentation occurs by focal, intermittent, nonpropulsive contractions of intestinal circular smooth muscle that repeatedly divide and mix luminal contents to enhance digestion and absorption. These segmentation contractions occur every few centimeters in the gut wall. Segmentation contractions chop and shear the contents in the gut, mixing them to maximize contact with the epithelium and digestive enzymes.

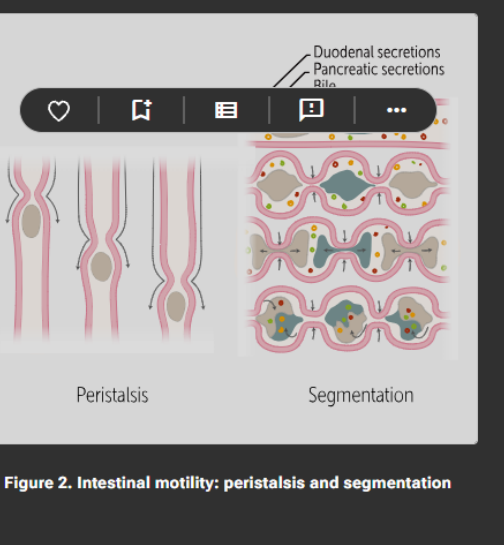

Figure 2. Intestinal motility: peristalsis and segmentation

CLINICAL REASONING

Motility disorders arise when this balance is disrupted. Too-slow motility allows contents to stagnate, leading to constipation, bacterial overgrowth, or ileus. Too-fast motility shortens epithelial contact time, contributing to diarrhea—though motility is rarely the sole mechanism. A fuller picture of diarrhea mechanisms appears after all six functions have been introduced, below.

The stomach deserves special mention: it functions as a reservoir, grinder, and gatekeeper, not merely a pump. Gastric emptying is tightly regulated by volume, caloric content, fat, acid load, and duodenal feedback signals, ensuring the small intestine is not overwhelmed.

Secretion

Secretion is the addition of fluids and chemical agents like acid, enzymes, bile, mucus, and water into the lumen of the GI tract in order to facilitate digestion and protect the mucosa. Note that secretion is not a one-way process: the ~7 liters of fluid secreted daily into the intestinal lumen are almost entirely reclaimed downstream—secretion and absorption together constitute the gut's fluid economy. The major secretory products include:

- **Gastric acid** (hydrochloric acid; HCl) is secreted from parietal cells in the stomach; this acid activates pepsinogen and kills ingested microbes.
- **Pepsinogen,** an inactive zymogen, is secreted from chief cells in the stomach. This is then cleaved by gastric acid (and autocatalytically by existing pepsin) to form the active enzyme pepsin which begins protein digestion in the stomach.
- **Mucus** is secreted by surface epithelial cells in the GI tract.
- **Pancreatic enzymes** (lipase, amylase, proteases) are secreted from acinar cells, and break down macromolecules (lipids, carbohydrates and proteins).
- **Bicarbonate** serves as the GI tract's primary acid-neutralizing buffer, secreted at three anatomically distinct sites: *gastric surface epithelial cells* (forming the mucus-bicarbonate barrier that protects the stomach lining), *Brunner glands in the duodenum* (neutralizing acid as chyme exits the stomach), and *pancreatic ductal cells* (delivering the largest bicarbonate volume into the duodenum to neutralize chyme). These three sites work in sequence from proximal to distal.
- **Bile salts** are synthesized in the liver, secreted into bile and stored and concentrated by the gallbladder. Bile serves to emulsify dietary fats in the duodenum. This makes the fats soluble in water, and facilitates their subsequent digestion

Digestion

Digestion is the process by which macromolecules (proteins, carbohydrates, and fats) are broken into absorbable units. It is important to note that digestion and absorption are not the same process. Digestion occurs via enzymatic and mechanical processes within the lumen and at the brush border of the small intestine, whereas absorption is the subsequent uptake of digested substances across the epithelium into the body. A failure of enzymatic digestion is properly called *maldigestion* rather than *malabsorption*—though the two are clinically linked, since undigested macromolecules cannot cross the epithelium and chronic maldigestion invariably leads to malabsorption downstream.

- **Proteins** are cleaved by pepsin (stomach) and pancreatic proteases, then by brush-border peptidases.
- **Carbohydrates** are broken by salivary and pancreatic amylase, then by brush-border disaccharidases like lactase and sucrase.
- **Fats** require bile salt emulsification—which disperses large fat droplets into smaller ones to dramatically increase surface area for enzymatic attack—before pancreatic lipase can hydrolyze triglycerides into fatty acids and monoglycerides for absorption. Most of the fat digestion occurs in the stomach, a small amount of fats are digested by gastric and lingual lipases.

Absorption

Absorption is the transfer of digested nutrients, water, and electrolytes from the lumen—technically still the external environment—into the blood or lymph; it is at this moment that ingested material truly enters the body. The small intestine, particularly the jejunum, is the primary absorptive site, with its surface area massivel microvilli (the brush border). Of note, lacteals are specialized structures within the villi that play a role in long chain fatty acid uptake and subsequent chylomicron production.

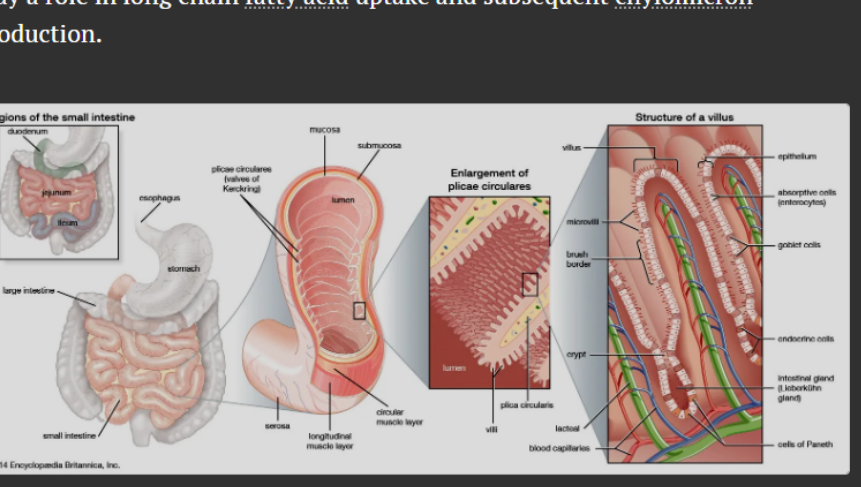
Regions of the small intestine
Structure of a villus
Enlargement of plicae circulares

Figure 3. Depiction of GI mucosal structures: plica circularis (large folds), villi, and microvilli

- **Glucose and galactose** are transported via sodium-glucose cotransporter 1 (SGLT1).
- **Amino acids** are transported via sodium-coupled transporters.
- **Long-chain fatty acids and monoglycerides** form micelles in the lumen, and at the brush border the lipid contents leave the micelles and cross the enterocytes apical membrane by diffusion and protein-mediated uptake. They are converted into triglycerides and packaged into chylomicrons and exported into the circulation via lymphatics (Figure 4).
- **Short-chain fatty acids** can be absorbed directly into the portal system without forming).
- **Most electrolyte and water** reabsorption occurs in the small intestine (specifically the jejunum and ileum). The large intestine (colon) reabsorbs the remaining water/electrolytes to concentrate feces.

CLINICAL CORRELATION

Steatorrhea—fat in the stool—represents a mass balance failure in fat absorption. Fat absorption requires four linked steps: bile salt emulsification of fat droplets, pancreatic lipase hydrolysis of triglycerides, micelle formation for mucosal uptake, and chylomicron packaging for lymphatic transport. Steatorrhea can result from defects in any of these steps. Detailed mechanisms are covered in a later brick.

CLINICAL REASONING

The portal circulation absorbs and delivers small, water-soluble nutrients to the liver for processing. In contrast, large lipoprotein particles such as chylomicrons cannot readily enter portal capillaries, so they are transported through the lymphatic system into the systemic circulation, where triglycerides are delivered to peripheral tissues such as muscle and adipose tissue. The portal system is covered in the Anatomy Blood Supply brick.

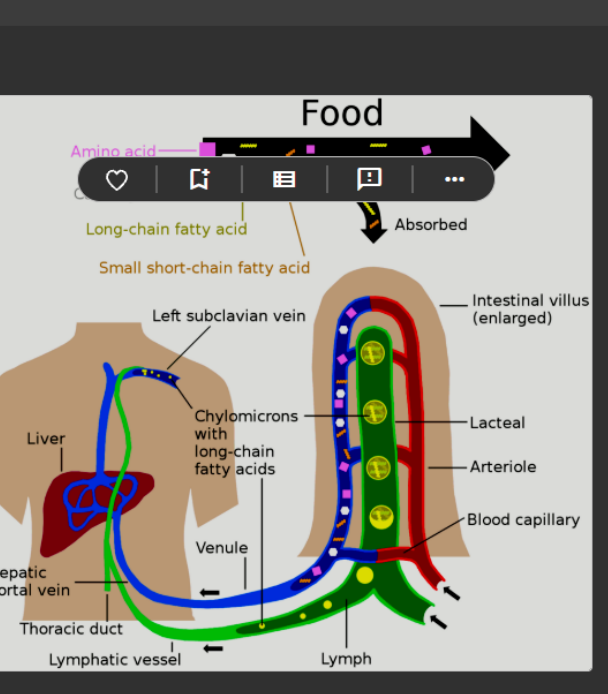

Figure 4. Absorption of amino acids, carbohydrates (capillaries to portal vein), and fatty acids (lacteals in the villi to lymphatics)

Protection

Protection allows the gut to digest food without digesting itself, tolerate the microbiome without uncontrolled inflammation, and absorb nutrients without absorbing pathogens. Because the lumen is continuous with the external world, every mucosal surface is in constant contact with potentially harmful material; the protective mechanisms below are what make controlled, selective interaction possible—rather than simple isolation.

- **Mucus layer** which forms a physical barrier.
- **Tight junctions** between epithelial cells are a paracellular barrier to movement of sol
- **Secretory IgA** is part of the gastrointestinal immune defense system and helps provide immune surveillance at mucosal surfaces.
- **Balance between gastric acid secretion and mucosal defense** is maintained by mucus and bicarbonate secretion, prostaglandin activity, and regulation of mucosal blood flow.

CLINICAL CORRELATION

Acid and pepsin, secreted by the stomach, are essential for protein digestion, but they are also capable of digesting the gastric mucosa. Normally, mucus, bicarbonate, prostaglandins, and mucosal blood flow form a defense layer (as you will learn more later).

Peptic ulcer disease (PUD) results when mucosal defenses are overwhelmed by injury. This can occur when *H. pylori* disrupts the mucus layer, NSAIDs inhibit prostaglandin-mediated protection, or less commonly, when excess gastrin (as in Zollinger-Ellison syndrome) causes true gastric acid hypersecretion. Understanding this balance explains why proton pump inhibitors (PPIs), H2 blockers, and prostaglandin analogs all have therapeutic roles.

Elimination

Elimination expels waste from the tract. Viewed through a mass balance lens, the colon manages the final stage of the GI fluid economy: under normal conditions, the small intestine delivers approximately 1–2 liters of fluid to the colon daily, and the colon reduces this to roughly 150 mL in stool—reclaiming over 90% of what it r n the small intestine, providing the prolonged contact time required for this salvage process. When transit is too slow, excess water is absorbed and constipation results. When fluid delivery exceeds the colon's reabsorptive capacity—whether from increased secretion, osmotic load, or inflammation upstream—excess water remains in the lumen and watery diarrhea results.

Defecation is a coordinated reflex involving rectal distension, relaxation of the internal anal sphincter (involuntary), and voluntary relaxation of the external anal sphincter.

CLINICAL CORRELATION

MECHANISMS OF DIARRHEA

Diarrhea is not a single failure—it is a mass balance disruption in which fluid delivery to the stool exceeds the colon's capacity to reclaim it. Now that all six GI functions have been introduced, consider how each can drive this imbalance:

Motility-related: Hypermotility accelerates transit and shortens epithelial contact time, limiting absorption. Motility dysfunction is rarely the sole driver and is most significant in combination with other defects.

Osmotic: Unabsorbed solutes remain in the lumen and draw water osmotically. Osmotic diarrhea characteristically resolves with fasting (removing the osmotic load). Classic example: lactase deficiency (JM). Lactase is a brush-border disaccharidase; deficiency leaves lactose in the lumen, causing osmotic diarrhea.

Secretory: Intestinal crypt cells hypersecrete Cl⁻ via CFTR, obligating water secretion into the lumen. Secretory diarrhea *persists with fasting*—fluid is driven into the lumen regardless of food intake. Classic example: cholera toxin activates adenylyl cyclase →

Inflammatory: Mucosal injury causes leakage of fluid, blood, proteins, and mucus into the lumen, disrupting both absorption and barrier protection. Classic example: Clostridioides difficile infection.

The three patients in this brick each illustrate a different mechanism. Returning to them after all six functions have been introduced—rather than mid-brick—allows the full diagnostic logic to land.

Regulation of Gut Function: Neural, Hormonal, and Paracrine Signals (LO 2)

Understanding what goes wrong in these six processes depends on understanding how they are controlled in the first place. The GI tract regulates itself through an integrated system of neural, hormonal, and paracrine signals—a system sophisticated enough that the gut has earned the nickname "the second brain." The GI tract is a mostly autonomously regulated system. It contains its own nervous system, responds to a suite of hormones, and uses local paracrine signals to fine-tune activity moment to moment.

Enteric Nervous System

The enteric nervous system (ENS) is a network of neurons embedded in the gut wall, organized into two main plexuses (Figure 5):

- The **myenteric plexus** (Auerbach's plexus), located between the circular and longitudinal muscle layers, primarily controls motility.
- The **submucosal plexus** (Meissner's plexus) regulates secretion and local blood flow.

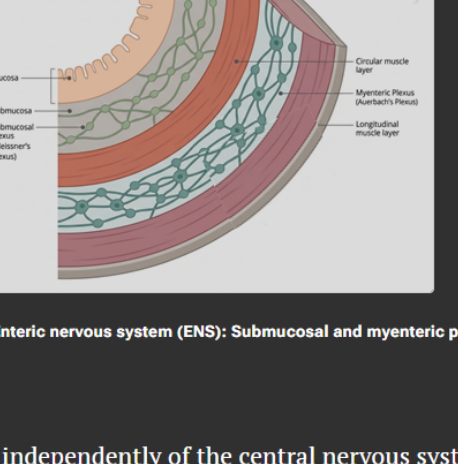

Figure 5. Enteric nervous system (ENS): Submucosal and myenteric plexi.

The ENS can operate independently of the central nervous system (CNS)—earning the gut its nickname as the "second brain." Local reflexes within the ENS coordinate peristalsis, segmentation, and secretory responses to luminal contents without requiring conscious input.

Mnemonic

Of the myenteric and submucosal plexuses, which nerve plexus controls which function?

Autonomic Nervous System (ANS) Modulation

The autonomic nervous system (ANS) modulates GI activity based on the body's overall state (fed, fasted).

Parasympathetic activity (PSNS) is associated with "rest and digest" or the fed state, and the release of acetylcholine (ACh). The PSNS generally increases GI activity—stimulating motility and secretion. This is primarily stimulated via the **vagus nerve** for the upper GI tract, and **pelvic nerves** for the distal colon.

Sympathetic input, released during stress or the "fight-or-flight" response, generally decreases GI activity by inhibiting motility and reducing secretion, while also constricting blood vessels. This explains why stress can cause constipation or slow gastric emptying, and why the gut becomes hyperactive when sympathetic tone is reduced.

Hormone and Paracrine Regulators

Hormones are released into the bloodstream and act at distant targets, whereas paracrine signals act locally near their site of release without significant entry into the systemic circulation. Examples of the major GI hormone and paracrine factors and their roles include:

- **Gastrin** is a hormone secreted in the antrum in response to protein and distension, and stimulates acid secretion and gastric motility.
- **Cholecystokinin** (CCK) is a hormone secreted in the proximal small intestine (duode nd protein. It stimulates pancreatic enzyme secretion and gallbladder contraction.
- **Secretin is a hormone** secreted in the duodenum in response to acid and stimulates pancreatic bicarbonate secretion.
- **Somatostatin** acts primarily in a paracrine fashion in the GI tract—though it also circulates as an endocrine signal in certain contexts. It is secreted by the stomach, duodenum, and pancreas, and functions as a broad inhibitory brake on acid secretion, gastrin release, and other GI functions.
- **Histamine** is a paracrine factor secreted in the gastric mucosa, and potently stimulates parietal cell acid secretion.

What are the two main categories of chemical regulators of GI function introduced here?

It is important to note that these factors play a role in **homeostatic control** mechanisms involving nervous, hormonal, and paracrine signals in positive and negative feedback. For example, and as you will learn in a later brick, regulated secretion o s of gastrin, histamine, and somatostatin as well as neuronal acetylcholine (ACh).

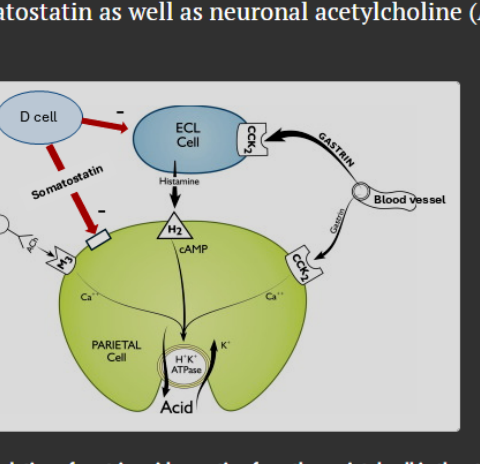

Figure 6. Regulation of gastric acid secretion from the parietal cell in the stomach

Example of Regulation of Gastric Acid Secretion

Gastric acid secretion illustrates how neural, hormonal, and paracrine signals converge on a single effector cell. Parietal cells receive three synergistic stimulatory inputs—acetylcholine (vagal), gastrin (endocrine), and histamine (paracrine via ECL cells)—that act in a feed-forward manner to drive acid output. Somatostatin provides counter-regulation by suppressing both parietal cells and ECL cells directly, functioning as true negative feedback. The full mechanistic detail of this pathway—including cephalic, gastric, and intestinal phases, and the clinical pharmacology of PPIs and H2 blockers—is covered in the dedicated gastric secretion brick.

Somatostatin provi ressing parietal cell acid secretion and by inhibiting histamine release from ECL cells. Somatostatin–mediated inhibition *is* true negative feedback.

Relating Dysfunction to Disease (LO 3)

Identifying which functional step is dysfunctional can be used systematically as a reasoning tool. Instead of memorizing disease facts in isolation, try to relate them to which GI function has failed. This approach is transferable to any GI disease you will encounter in clinical training. The table below maps each core GI process to its failure pattern and representative diseases:

Table 2. Linking GI dysfunction to diseases (You will learn these diseases later. We wanted to make the link between function, dysfunction, and disease early on)

Function	Failure of Function	Example Diseases with dysfunction
Motility	Too fast or too slow transit	Achalasia, ileus, constipation, diarrhea
Secretion	Excess or absent secretory products	Cholera, Zollinger-Ellison syndrome
Digestion	Enzyme deficiency	Pancreatic insufficiency, lactase deficiency
Absorption	Mucosal injury or transporter loss	Celiac disease, short bowel syndrome
Protection	Barrier breakdown	Peptic ulcer disease, *H. pylori* infection, GERD
Elimination	Transit or defecation reflex	Constipation, Hirschsprung

† *Note: Hirschsprung disease involves colonic aganglionosis—absent ENS ganglia cause a motility failure that manifests as failure of elimination. Instructors may prefer to list it under Motility depending on the level of mechanistic detail desired for their course.*

Common Misconceptions

Several persistent misconceptions derail GI reasoning early in medical training. Explicitly correcting these now will prevent errors later. The most important ones to internalize are:

1. **"The stomach is the main site of digestion and absorption."** False. The stomach initiates protein digestion and acidifies chyme, but absorbs almost nothing. The jejunum is the primary absorptive site. This misconception leads students to mislocalize the consequences of gastric disease.
2. **"Diarrhea means the gut is moving too fast."** Incomplete. Hypermotility is one contributor, but most clinically significant diarrhea is driven by secretory, osmotic, or inflammatory mechanisms—not motility alone. Treating diarrhea purely by slowing transit (e.g., loperamide) can be ineffective or harmful if the underlying mechanism is not addressed.
3. **"Maldigestion and malabsorption are the same thing."** False. Maldigestion is failure to break macromolecules down enzymatically; malabsorption is failure to transport digested products across the epithelium. They have different causes (enzyme deficiency vs. mucosal injury) and different clinical presentations—though chronic maldigestion leads to malabsorption downstream.
4. **"Bile absorbs fa** break large droplets into smaller ones to increase surface area for pancreatic lipase. Absorption itself requires lipase hydrolysis, micelle formation, and epithelial transport. Bile is necessary but not sufficient for fat absorption; failure at any subsequent step still causes steatorrhea. Note, bile salts are reabsorbed in the ileum.

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to TL, RK, and JM—do all three patients have the same underlying problem? The answer is no, and working through which function is "broken" makes this clear.

TL, the traveler with watery diarrhea, most likely has secretory diarrhea: an enterotoxin (such as from enterotoxigenic E. coli) has triggered crypt cells to hypersecrete Cl⁻ and water, overwhelming the colon's reabsorptive capacity. The failed process is secretion—specifically, dysregulated ion transport.

RK, with greasy stools from chronic pancreatitis, has a digestion failure: loss of pancreatic lipase means dietary fat cannot be hydrolyzed into absorbable units. The mass balance consequence is visible in the stool—fat that cannot be absorbed must be excreted, producing steatorrhea. The failed process is digestion; the mass balance outcome is elimination of what should have been absorbed.

JM's post-milk diarrhea is an osmotic mechanism: lactase deficiency leaves undigested lactose in the lumen, which draws water osmotically.

Same symptom—diarrhea—three entirely different failed processes. GI physiology gives you the tools to tell them apart.

Summary

- The GI tract is a regulated tube with six coordinated functions—Move, Secrete, Digest, Absorb, Protect, and Eliminate—and the lumen is technically outside the body until absorption occurs across the epithelial barrier.
- Each GI region is specialized: the jejunum is the primary absorption site, the terminal ileum recovers bile salts and B12, and the colon's main job is water and electrolyte salvage, not nutrient absorption.
- Digestion and absorption are distinct processes: digestion breaks macromolecules into absorbable units in the lumen; absorption transfers those units across the epithelium into blood or lymph.
- Acid secretion control is not regulated via a single pathway; ACh, gastrin, histamine, and somatostatin all contribute.
- The stomach is not the main absorption site: the jejunum handles most nutrient absorption.
- Bile is not an enzyme, but a detergent that emulsifies fat, making it accessible for digestion by lipase.
- GI function is controlled by three integrated layers: the enteric nervous system (local reflexes), the autonomic nervous system (parasympathetic increases, sympathetic decreases activity), and hormones/paracrines (gastrin, CCK, secretin, histamine, somatostatin).
- Diarrhea is not simply increased motility; it results from secretory, osmotic, inflammatory, or motility-related mechanisms, each pointing to a different failed physiological process.
- Steatorrhea requires identifying the specific failed step in fat handling: bile salt deficien ry, or lymphatic obstruction. Each produce the same symptom through different mechanisms.
- A stepwise framework—identify the symptom, map it to a failed GI process, localize the region, and consider the control system—converts GI symptoms into mechanistic diagnoses and is applicable to any GI disease.

Review Questions

1. Which core GI process breaks macromolecules into absorbable units?

A. ○ Motility
B. ● Digestion
C. ○ Absorption
D. ○ Secretion
E. ○ Protection

Hide Explanation

Correct answer (B) Digestion breaks down macromolecules in the lumen and brush border. (A) Motility moves contents; (C) Absorption transfers units across epithelium; (D) Secretion adds digestive fluids; (E) Protection prevents self-digestion. Thus, only (B) directly refers to breaking down macromolecules.

2. Chronic pancreatitis involves greasy, foul-smelling stools. Which GI process has primarily failed?

A. ○ Motility failure
B. ● Digestion failure
C. ○ Absorption failure
D. ○ Secretion failure
E. ○ Protection failure

Hide Explanation

The correct answer is Digestion (B). Loss of pancreatic lipase means dietary fat cannot be hydrolyzed into absorbable units, causing steatorrhea—a digestion failure. Motility (A) refers to propulsion and mixing of contents, not enzymatic breakdown. Absorption (C) transfers already-digested units across the epithelium; fat never reaches this step. Secretion (D) adds fluids and enzymes to the lumen; while pancreatic secretion is impaired, the core functional failure is digestion itself. Protection (E) prevents mucosal self-injury and is not the primary defect here.

3. Watery diarrhea occurs when enterotoxins activate CFTR chloride channels, increasing chloride and water in the intestinal lumen. Which GI process has primarily failed?

A. ○ Motility failure
B. ● Secretion failure
C. ○ Digestion failure
D. ○ Absorption failure
E. ○ Protection failure

Hide Explanation

Correct answer (B) Secretion failure. Enterotoxins dysregulate ion transport, causing hypersecretion that overwhelms colonic reabsorption. Motility failure (A) involves abnormal propulsion or mixing, not ion transport. Digestion failure (C) reflects enzyme deficiency breaking down macromolecules. Absorption failure (D) involves impaired epithelial transport of already-digested nutrients. Protection failure (E) reflects barrier breakdown, not secretory dysregulation.

4. The wave-like contractions in the small intestine are primarily coordinated by cells located in which of the following anatomical layers?

A. ● Between the circular and longitudinal muscle layers
B. ○ Within the mucosa next to the epithelium
C. ○ In the submucosa surrounding blood vessels
D. ○ Within the serosa covering the outside of the intestine

Hide Explanation

Correct answer (A). The myenteric (Auerbach's) plexus is located between the 2 muscle layers, containing enteric neurons that control the peristaltic contractions. As you will learn later, the interstitial cells of Cajal (ICC) in the muscularis externa (near the myenteric plexus) generate slow waves; the myenteric plexus coordinates motility. Incorrect answers (B). This is the region plays a role in secretion and blood flow. (C) The submucosa regulates secretion and local blood flow. (D) The serosa is a connective tissue on the outside of the GI tract.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

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02 ☑ ~~Compa n by the autonom s system (ENS),~~

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Objective Confidence: 3/3